

Shang Ma

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EDUCATION

University of Science and Technology of China Hefei, China
Ph.D. Candidate in Geophysics **2021 - Present**
Advisor: Zefeng, Li (zefengli@ustc.edu.cn)

University of Science and Technology of China Hefei, China
B.S. in Geophysics **2017 - 2021**

RESEARCH INTERESTS

Machine Learning Seismology: Applying machine learning to uncover low-SNR seismic signals (e.g., PKP precursors), map deep mantle structures, and analyze earthquake sources.

RESEARCH EXPERIENCE

Global & Regional PKP Precursor Analysis Apr., 2025 - Present
Classified coherent versus incoherent PKP precursors automatically.
Applied GraphCursor to systematically detect PKP precursors recorded by China National Seismograph Network (CNSN).
Performed Phonon simulations to estimate the scattering strength of incoherent PKP precursors recorded by CNSN, revealing values slightly lower than global averages.

Global search of PKP precursor with graph neural networks Aug., 2024 - Aug., 2025
Developed GraphCursor, a Graph Neural Network integrating multistation waveforms and locations to robustly detect low-SNR PKP precursors.
Applied migration techniques to locate scatterers in the lowermost mantle and quantify scattering strength, producing a global map of small-scale scatterers.

Deciphering deep mantle hemispheric geochemical dichotomy Dec., 2022 - Jun., 2024
Used random forests to test the credibility of LLSVP-based versus hemispheric mantle zoning across multiple global oceanic island basalts (OIBs) isotopic datasets.
Found strong support for hemispheric zoning, with dataset-dependent discrepancies for the LLSVP dichotomy.

Machine learning of source spectra for large earthquakes Mar., 2021 - Jun., 2022

Applied variational autoencoder to build single- and double-variable models of earthquake source spectra from global events, improving fit by ~8.5% over the Brune model.

Developed double-variable models capturing second-order spectral features, providing insights to refine physics-based source models.

PUBLICATIONS

5. **Ma, S.**, Li, Z., Wang, W., Sun, D., Li, J., & Su, Y. (*under review*). Systematic Detection of PKP Precursors on China National Seismograph Network with Deep Learning.
4. **Ma, S.**, Li, Z., Chen, L., Shen, J., Li, Y., Wang W., & Len, W. (*in revision*). Deciphering Earth's deep mantle hemispheric geochemical dichotomy with machine learning.
3. **Ma, S.**, Li, Z., Sun, D., Su, Y., Li, J., Si, X., & Zhu, J. (2025). Global Search of PKP Precursors With Graph Neural Network: Implications for Scatterers in the Lowermost Mantle. *Geophysical Research Letters*, 52(17), e2025GL115952. <https://doi.org/10.1029/2025GL115952>
2. Cui, X., Hu, Y., **Ma, S.**, Li, Z., Liu, G., & Huang, H. (2024). Bridging Supervised and Unsupervised Learning to Build Volcano Seismicity Classifiers at Kilauea Volcano, Hawaii. *Seismological Research Letters*, 95(3), 1849 - 1857. <https://doi.org/10.1785/0220230251>
1. **Ma, S.**, Li, Z., & Wang, W. (2022). Machine learning of source spectra for large earthquakes. *Geophysical Journal International*, 231(1), 692-702. <https://doi.org/10.1093/gji/ggac215>

PRESENTATIONS

Systematic detection of PKP precursors on China National Seismograph Network with deep learning

Annual Meeting of Chinese Geoscience Union **Oct., 2025**

Detecting PKP precursors using graph neural networks: implications for scatterers in the lowermost mantle

American Geophysical Union Annual Meeting **Dec., 2024**

Deciphering Earth's deep mantle hemispheric geochemical dichotomy with machine learning

Youth Scientist Forum Of Earth Science **May., 2024**

Machine learning of source spectra for large earthquakes

Seminar on AI and microseismicity monitoring **May., 2021**

HONORS & AWARDS

National Scholarship, Ministry of Education, China **2025**

Special Talent Cultivation Program for PhD Students, China Association for Science and Technology **2025**

Outstanding Student Presentation Award, Youth Scientist Forum Of Earth Science **2024**

Outstanding Poster Award, International Professional Association of Chinese Earth Scientists **2023**

First-Class Graduate Scholarships, USTC **2021-2025**

Kwang-Hua Scholarship, USTC **2020**

Outstanding Student Scholarship, USTC **2019**